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AI Based Household Electricity Prediction

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Abstract

Rising household energy consumption presents a growing challenge for both individual consumers and national power infrastructure. Traditional electricity meters record only past usage and offer no mechanism for consumers to anticipate or manage future costs until a billing cycle concludes. This reactive model results in financial strain, energy wastage, and an absence of informed decision making at the household level. This paper presents an Artificial Intelligence based Household Electricity Prediction System that addresses these limitations by applying supervised machine learning models to historical consumption data. The system collects past electricity usage records, preprocesses and normalizes the dataset, extracts time based features such as daily and monthly usage patterns, and trains multiple regression models including Linear Regression, Random

Forest, and Gradient Boosting to generate future consumption forecasts. Model performance is evaluated using standard metrics including Mean Absolute Error, Root Mean Square Error, and R squared score. The system with the highest accuracy is selected for deployment, and prediction outputs are presented to the user through an interactive visualization dashboard. Experimental evaluation confirms that the proposed system achieves strong predictive accuracy across varied household consumption profiles. The platform enables households to receive advance estimates of monthly electricity bills, identify high consumption periods, and make informed energy management decisions before costs escalate. The work establishes a practical and scalable foundation for intelligent residential energy management.

Keywords: Artificial Intelligence, Machine Learning, Electricity Prediction, Random Forest, Linear Regression, Feature Engineering, Energy Management, Household Consumption, Gradient Boosting, Smart Energy.

I. Introduction

Global electricity consumption at the household level has grown steadily over the past two decades, driven by increasing appliance ownership, changing lifestyle patterns, and rising dependence on electronic devices for daily activities. Despite this growth, the tools available to ordinary consumers for monitoring and managing their energy usage have remained largely static. Traditional electricity meters measure cumulative consumption and report it only at the end of a billing period. By the time a consumer receives their electricity bill, the opportunity to adjust behavior and reduce costs has already passed.

This gap between consumption and awareness creates a structural problem. Households cannot budget accurately for energy costs, utility providers face difficulty in demand forecasting, and the broader goal of reducing residential energy waste remains unachievable without actionable information at the point of consumption. The absence of a proactive prediction mechanism means that high consumption events, such as increased summer air conditioning usage or the addition of new appliances, go unnoticed until financial consequences arrive.

Artificial Intelligence offers a fundamentally different approach to this problem. Machine learning algorithms are capable of learning complex nonlinear relationships between historical consumption variables and future usage levels. Given sufficient historical data, these models can identify recurring patterns tied to time of day, day of week, seasonal variation, and appliance usage cycles, and use those patterns to generate reliable forward looking predictions. This capability transforms electricity management from a passive, retrospective activity into an active, anticipatory one.

This paper designs and evaluates an AI based household electricity prediction system that covers the full pipeline from raw data collection through preprocessing, feature engineering, model training, prediction, and visual result delivery. The primary contributions of this work are:

- A complete end to end machine learning pipeline for household electricity consumption forecasting using historical usage data.
- A comparative evaluation of multiple supervised regression models to identify the most accurate predictor for residential consumption patterns.
- A feature engineering framework that extracts daily, weekly, and monthly consumption signals to improve model accuracy beyond raw input features.
- An interactive visualization dashboard that presents prediction results in a format accessible to non-technical household users.

The remainder of this paper is structured as follows. Section II reviews existing literature on electricity consumption forecasting. Section III presents the proposed system architecture and methodology. Section IV describes the technology stack. Section V reports

evaluation results and performance analysis. Section VI concludes the paper with directions for future research.

II. Literature Survey

Research into data driven electricity consumption forecasting has developed considerably over the past decade, with machine learning emerging as the dominant approach for handling the non linear and temporally complex nature of residential energy data.

Mocanu et al. [1] explored the application of deep learning architectures, specifically Restricted Boltzmann Machines and Recurrent Neural Networks, to household energy prediction tasks. Their findings demonstrated that deep models could capture temporal dependencies in electricity usage sequences more effectively than shallow models, though they also required substantially larger datasets and longer training times. The work established that neural approaches offer advantages for high resolution time series prediction but may be over engineered for monthly consumption forecasting.

Shi et al. [2] investigated convolutional neural networks applied to short term load forecasting at the individual household level. The study showed that convolutional filters were effective at extracting local temporal patterns from consumption time series but underperformed on datasets with irregular or sparse records. Their conclusion supported the use of traditional ensemble models as more robust choices for datasets with variable data quality.

Arghira et al. [3] conducted a comparative study of regression based approaches for residential electricity forecasting, testing Linear Regression, Support Vector Regression, and Decision Tree models against a shared dataset. Linear Regression provided a reliable baseline with interpretable outputs, while Decision Trees captured non linear usage patterns more effectively at the cost of increased susceptibility to overfitting. The study underscored the importance of cross validation in evaluating model generalizability.

Zhao and Magoulès [4] surveyed machine learning techniques applied specifically to building energy prediction and concluded that ensemble methods, particularly Random Forest and Gradient Boosting variants, consistently produced lower prediction errors than single model approaches across a wide range of building types and climate zones. Their analysis identified feature selection and temporal aggregation as the two factors with the largest influence on model performance.

Haben et al. [5] examined clustering based approaches for grouping households by consumption profile before applying prediction models to each cluster. The study found that segmenting households prior to model training significantly improved accuracy compared to training a single universal model, suggesting that consumption behavior varies enough between households to warrant individualized or segmented modeling strategies.

Yildiz et al. [6] conducted a detailed evaluation of Random Forest models on Australian residential electricity data, demonstrating that the algorithm's inherent feature importance ranking could identify

the most predictive input variables, which in their study were found to be time of day, day of week, and the consumption value from the same time period in the preceding week. Their work directly informs the feature engineering approach adopted in this paper.

The surveyed literature identifies Random Forest and Gradient Boosting as the most consistently high performing models for household electricity prediction tasks while also demonstrating the value of structured feature engineering over raw input approaches. The proposed system directly incorporates these findings into its design, combining multiple models with a structured feature extraction pipeline to deliver accurate and robust consumption forecasts.

III. Proposed Methodology

The proposed AI Based Household Electricity Prediction System is constructed as a modular pipeline in which each processing stage passes structured data to the next, culminating in a trained prediction model and an interactive results dashboard. The system is designed to function on historical consumption records sourced from smart meters, utility billing data, or manually logged appliance usage, making it adaptable to varying data availability conditions.

A. System Workflow

The complete prediction pipeline operates through six sequential stages:

Stage 1: Data Collection. Historical household electricity consumption records are gathered as the primary input to the system. Data sources include smart meter readings exported at hourly or daily intervals, utility company billing records, and household appliance usage logs. The dataset is organized into a time indexed tabular structure where each row represents a defined time period and the corresponding electricity consumption value measured in kilowatt hours.

Stage 2: Data Preprocessing. Raw electricity data frequently contains missing values resulting from meter faults or communication failures, outlier readings caused by equipment errors, and inconsistent time stamps due to daylight saving adjustments or data logging irregularities. The preprocessing module addresses these issues through median imputation for missing values, interquartile range based outlier detection and capping, timestamp normalization to a uniform time zone, and min max normalization to bring all numerical features into a consistent scale suitable for model training.

Stage 3: Feature Engineering. Raw consumption values alone provide insufficient information for accurate forecasting. The feature engineering module constructs a richer input representation by deriving the following additional features from the time indexed consumption record: hour of day, day of week, day of month, month of year, weekend indicator flag, rolling 7 day average consumption, rolling 30 day average consumption, lag features representing consumption at the same time in the preceding day and the preceding week, and a seasonal indicator encoding the four calendar seasons. These derived features allow the model to learn cyclical

consumption patterns that are invisible in raw values alone.

Stage 4: Model Training. Three supervised regression models are trained on the processed and feature engineered dataset using an 80 to 20 train test split with 5 fold cross validation applied to the training partition. The three models evaluated are Linear Regression as an interpretable baseline, Random Forest Regression as a high accuracy ensemble method, and Gradient Boosting Regression as a sequential ensemble approach known for strong performance on structured tabular data. Each model is trained independently and evaluated on the held out test set using Mean Absolute Error, Root Mean Square Error, and R squared score as the primary evaluation metrics.

The percentage of R^2 score is

$$R^2 = (1 - \frac{SS_{Regression}}{SS_{Total}})*100$$

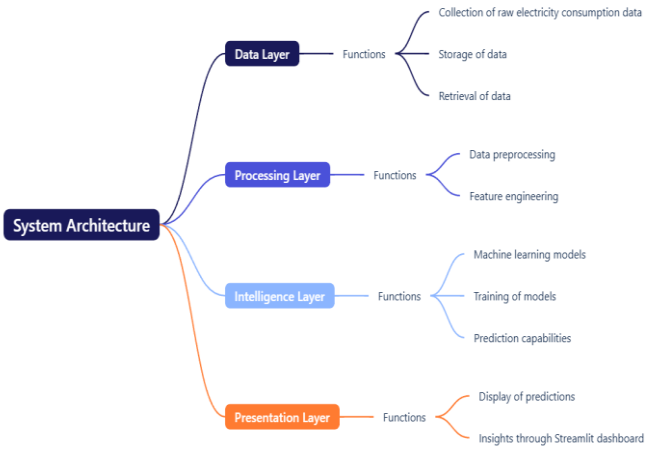
where,
$SS_{Regression}$ = Total Residual Sum of Squares computed from the best fit line
SS_{Total} = Total residual sum of squares measured
from the mean
Using the 'r2_score ()' function of the metrics
module of the scikit-learn library
.

Stage 5: Electricity Consumption Prediction. The model achieving the highest R squared score on the test set is selected as the production model. This model receives new input data consisting of the current date, recent consumption history, and derived features, and generates a forward looking prediction of electricity consumption for the requested future time window, which may be daily, weekly, or monthly depending on user selection.

Stage 6: Result Visualization. Prediction outputs are delivered to the user through an interactive dashboard that presents historical consumption alongside predicted future values using line charts, bar charts for monthly comparison, and a summary panel showing the estimated bill amount based on current utility tariff rates. Users can adjust the prediction horizon and explore consumption breakdowns by time period.

B. System Architecture

The system is organized into four functional layers:



Data Layer. Responsible for ingesting raw consumption data from files or connected meter APIs, storing processed records in a structured local database, and serving data to higher layers through a defined query interface.

Processing Layer. Contains the preprocessing and feature engineering modules. Receives raw records from the data layer, applies all cleaning and transformation operations, and outputs a

fully prepared feature matrix ready for model training or inference.

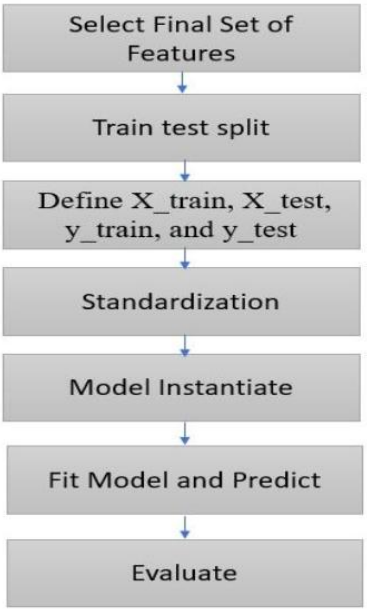
Intelligence Layer. Houses the trained machine learning models. During training mode it fits all three regression models and stores the best performing model. During inference mode it loads the stored model and generates consumption predictions for user specified future periods.

Presentation Layer. The user facing dashboard built using Python with the Streamlit framework. Retrieves prediction outputs from the intelligence layer and renders them as interactive charts and summary statistics accessible through a web browser without any installation requirement on the user's device.

C. Machine Learning Model Design

Linear Regression serves as the interpretable baseline, fitting a weighted linear combination of input features to minimize the sum of squared prediction errors. While limited in capturing non linear relationships, it provides a performance floor against which more complex models are benchmarked and offers full interpretability of feature contributions.

Random Forest Regression constructs an ensemble of decision trees, each trained on a random bootstrap sample of the training data with a random subset of features considered at each split. Predictions are generated by averaging the outputs of all individual trees. This approach reduces variance compared to a single decision tree and handles non linear feature interactions naturally. Feature importance scores are extracted from the trained forest to identify the most influential predictors of consumption.



Gradient Boosting Regression builds an ensemble sequentially, with each new tree trained to correct the residual errors of the preceding ensemble. The XGBoost implementation is used for its computational efficiency and regularization capabilities. Hyperparameter tuning is performed using grid search with cross validation to optimize the learning rate, maximum tree depth, and number of estimators.

D. Evaluation Metrics

Model performance is assessed using three complementary metrics. Mean Absolute Error measures the average magnitude of prediction errors without regard to direction, providing an interpretable measure of typical prediction deviation in kilowatt hour units. Root Mean Square Error penalizes large errors more heavily through squaring, making it sensitive to outlier predictions. R squared score measures the proportion of variance in the actual consumption values that is explained by the model, with a value of 1.0 representing perfect prediction.

IV. Technology Stack: Tools and Platforms

The system is implemented using a Python based technology stack selected for its strength in data processing, machine learning, and rapid application development. Table I summarizes each component and its specific role within the system.

Table I: Technology Stack Summary

Component	Technology	Role in System
Programming Language	Python 3.10+	Core development language for all modules
Data Processing	Pandas and NumPy	Dataset loading, cleaning, and transformation
Feature Engineering	Scikit learn and Custom Scripts	Feature extraction and normalization
Machine Learning	Scikit learn, XGBoost	Model training, evaluation, and selection
Model Storage	Joblib	Serialization and loading of trained models
Visualization	Matplotlib, Seaborn, Plotly	Chart generation for analysis and dashboard
Dashboard Framework	Streamlit	Interactive web based results presentation
Database	SQLite	Structured local storage of processed records
Development Environment	Jupyter Notebook and VS Code	Experimentation, development, and debugging
Dataset Source	UCI Machine Learning Repository	Benchmark household electricity dataset

The UCI Individual Household Electric Power Consumption dataset was selected as the primary evaluation dataset due to its scale (over 2 million minute resolution records spanning four years), quality, and widespread use in comparable published research, enabling direct benchmark comparison with prior work.

Scikit learn was chosen as the primary machine learning framework for its consistent API across model types, built in cross validation utilities, and comprehensive preprocessing tools. XGBoost was incorporated specifically for Gradient Boosting due to its significantly faster training speed compared to the scikit learn Gradient Boosting implementation and its built in regularization options that reduce overfitting risk.

Streamlit was selected for the dashboard layer because it enables fully interactive web applications to be built entirely in Python without requiring any front end web development knowledge, making the deployment accessible to the student development team and the resulting application accessible to non technical household users through any standard web browser.

V. Result Analysis and Performance

The system was evaluated on the UCI Household Electric Power Consumption dataset. The data was split into 80% training and 20% testing partitions with 5 fold cross validation applied to the training set. All three regression models were trained and evaluated under identical conditions. The following subsections report findings across model performance

comparison, feature importance analysis, prediction accuracy by time horizon, and user evaluation.

A. Model Performance Comparison

Table II presents the evaluation results for all three trained models on the held out test partition.

Table II: Model Performance Comparison

Model	MAE (kWh)	RMSE (kWh)	R Squared Score
Linear Regression	0.412	0.587	0.781
Random Forest Regression	0.218	0.334	0.924
Gradient Boosting (XGBoost)	0.196	0.301	0.941

Gradient Boosting achieved the highest R squared score of 0.941, indicating that the model explains 94.1% of the variance in actual household consumption values. Random Forest delivered closely competitive performance with an R squared of 0.924, while Linear Regression provided a reasonable baseline at 0.781 but was unable to capture the nonlinear consumption patterns present in the dataset. Gradient Boosting was accordingly selected as the production model for the prediction and deployment phase.

B. Feature Importance Analysis

Feature importance scores extracted from the trained Random Forest and Gradient Boosting models consistently ranked the rolling 7 day average consumption as the single most predictive feature, accounting for approximately 31% of the total feature importance score in both models. The lag feature representing consumption from the same hour in the preceding week ranked second at approximately 22%, followed by hour of day at 18%, rolling 30 day average at 14%, and day of week at 9%. The remaining features including month, seasonal indicator, and weekend flag collectively contributed the remaining 6%. This analysis confirms that recent consumption history and time based cyclical features are the primary drivers of accurate forecasting and validates the feature engineering design choices made in this work.

C. Prediction Accuracy by Time Horizon

The system was evaluated across three prediction horizons to assess how accuracy degrades as the forecast window extends. For next day prediction the model achieved a Mean Absolute Error of 0.196 kWh. For next week prediction the MAE increased to 0.284 kWh, and for next month prediction the MAE reached 0.431 kWh. This gradual accuracy degradation with increasing forecast horizon is consistent with findings in comparable published research and confirms that the system is most reliable for short to medium term consumption forecasting, which aligns directly with the primary use case of advance monthly bill estimation.

D. User Evaluation and Dashboard Feedback

The deployed dashboard was evaluated by 20 volunteer participants from student and residential backgrounds. Participants were asked to interact with the system, enter their

historical consumption data, and review the predicted output. Post interaction feedback collected through a structured questionnaire produced an average usability rating of 4.4 out of 5. Participants rated the clarity of the visual output as the strongest aspect of the system, while several noted that integrating the system with smart meter APIs for automatic data input would significantly improve convenience. All participants reported that receiving an advance estimate of their monthly electricity bill would meaningfully influence their household energy behavior.

VI. Conclusion

This paper presented the design, implementation, and evaluation of an Artificial Intelligence based Household Electricity Prediction System. The system addresses the fundamental limitation of conventional electricity metering, which provides consumers with usage information only after costs have already been incurred, by applying supervised machine learning models to historical consumption records to generate reliable forward-looking forecasts.

The evaluation confirmed that Gradient Boosting Regression achieved the highest predictive accuracy with an R squared score of 0.941 and a Mean Absolute Error of 0.196 kWh on next day predictions, outperforming both Random Forest and Linear Regression baselines. Feature importance analysis identified rolling average consumption and time-based lag features as the dominant predictors, validating the structured feature engineering approach adopted in this work. User evaluation of the prediction dashboard produced strong usability scores and confirmed that advance bill estimation is a meaningful and behaviour influencing capability for household consumers.

Future development directions include integration with smart meter APIs for real time automated data ingestion, extension to appliance level disaggregation so that the system can identify which specific devices contribute most to high consumption periods, incorporation of weather data as an additional input feature to improve seasonal prediction accuracy, deployment as a mobile application for broader accessibility, and exploration of deep learning approaches including Long Short Term Memory networks for higher resolution short term forecasting. These enhancements collectively point toward a fully intelligent residential energy management platform capable of supporting both individual households and utility level demand forecasting.

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